Constraints on Higgs-charm couplings at CMS

Chirayu Gupta
on behalf of the CMS Collaboration

IIHE, Vrije Universiteit Brussel (VUB)

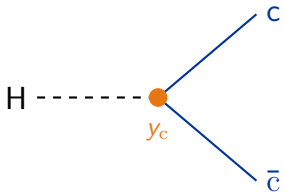
Higgs 2026, Louvain-la-Neuve, 28 October 2026

Why the Higgs-charm coupling?

- Higgs couplings to the 3rd generation (t, b, τ) and to μ are established
- **2nd-generation quark Yukawas are the open frontier**: charm is the most accessible
  - SM: $\mathcal{B}(\mathrm{H} \to \mathrm{c\bar{c}}) \approx 2.9\%$, about 20× below $\mathrm{H} \to \mathrm{b\bar{b}}$
  - Huge QCD backgrounds, c-jets hard to tell apart from b- and light jets
- BSM: flavour-compatible models allow $|\kappa_\mathrm{c}|$ of order 3

  e.g. spontaneous flavour violation, vector-like quarks (arXiv:2410.05236)

  $\Rightarrow$ current sensitivity is now entering this region
- Is the Higgs mechanism also responsible for the 2nd-generation quark masses?



Coupling modifier

$$\kappa_\mathrm{c} = y_\mathrm{c} / y_\mathrm{c}^\mathrm{SM}$$

(κ framework; SM: $\kappa_\mathrm{c} = 1$)

How CMS probes κ_c

Direct $H \to c\bar{c}$ (rate $\propto \kappa_c^2/\Gamma_H$)

VH($H \to c\bar{c}$)	PRL 131 (2023) 061801
$t\bar{t}H(H \to c\bar{c})$ + VH comb.	PRL 136 (2026) 011801
Boosted $H \to c\bar{c}$ (mostly ggH)	PRL 131 (2023) 041801

best: $|\kappa_c| < 3.5$ (2.7 exp.)

H + charm production (cg $\to$ cH)

cH($H \to \gamma\gamma$)	JHEP 11 (2025) 060
cH($H \to WW$) + comb. **NEW**	JHEP 08 (2026) 051

best: $|\kappa_c| < 47$ (51 exp.)

Indirect: H kinematics / production

p_T^H spectra **NEW**	PLB 792 (2019); JHEP 04 (2026)
γH + on-shell $H \to 4\ell$	PRD 112 (2025) 112001
Off-shell + on-shell $H \to 4\ell$ **NEW**	arXiv:2607.23352

$\kappa_c \in [-3.7, 3.2]$ to $[-8.6, 7.5]$ (by assumptions)

Rare exclusive decays

$H \to J/\psi\gamma$, $\psi(2S)\gamma$ PLB 865 (2025) 139462

sensitive to the *sign* of κ_c (interference)

$-166 < \kappa_c < 208$

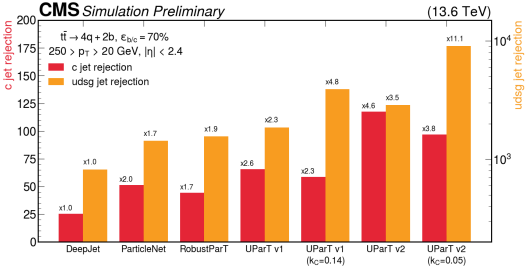
All results with 13 TeV Run 2 data, up to 138 fb^{-1}. **NEW** = public since Higgs 2025.

The key tool: charm-jet identification

- c-jets sit between b and light jets: shorter lifetime, fewer displaced tracks, softer leptons
- Taggers must separate **c vs. light** and **c vs. b** at the same time
- Rapid progress through ML on jet constituents:

  DeepJet → ParticleNet → (Robust)ParT → **UParT**

  - Several-fold gain in c-jet rejection over the Run 2 taggers at similar efficiency
- Merged-jet (boosted) $\mathrm{H} \to \mathrm{c\bar{c}}$ tagging with ParticleNet / GloParT
- Calibration in data (W+c, $\mathrm{t\bar{t}}$ semileptonic, Z+jets) is a leading systematic



CMS-DP-2025-081: c- and light-jet rejection by tagger generation (13.6 TeV simulation)

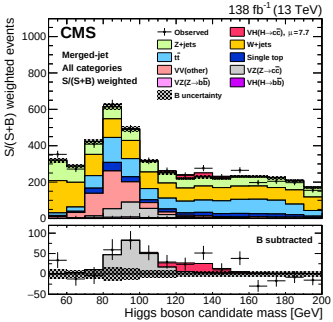
VH($\mathrm{H} \to \mathrm{c\bar{c}}$): the benchmark channel

- W/Z leptonic decays (0, 1, 2 leptons) for triggering and background suppression
- **Resolved** (two AK4 c-jets) + **merged** (one AK15 jet, ParticleNet) topologies
- Validated with VZ($\mathrm{Z} \to \mathrm{c\bar{c}}$): **first observation at a hadron collider**, 5.7σ

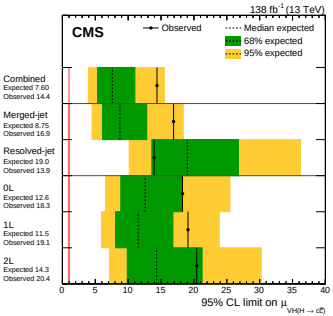
Result, $138\,\mathrm{fb}^{-1}$

$\mu_{\mathrm{VH(cc)}} < 14\,(7.6)$

$1.1 < |\kappa_{\mathrm{c}}| < 5.5 \quad (|\kappa_{\mathrm{c}}| < 3.4 \text{ exp.})$



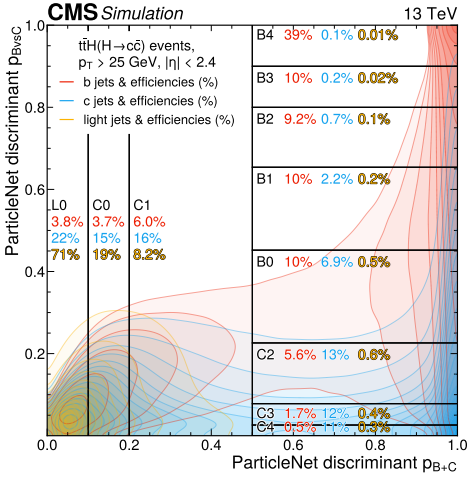
PRL 131 (2023) 061801, CMS-HIG-21-008



$\mathrm{t\bar{t}H(H \to c\bar{c})}$: a new channel

PRL 136 (2026) 011801, CMS-HIG-24-018

- First search for $\mathrm{H \to c\bar{c}}$ in $\mathrm{t\bar{t}H}$; 0, 1, 2 lepton $\mathrm{t\bar{t}}$ decays
- Many jets of all flavours: ParticleNet b vs. c vs. light **2D categories**
- ParT event classifier separating $\mathrm{t\bar{t}H(cc)}$, $\mathrm{t\bar{t}H(bb)}$, $\mathrm{t\bar{t}Z}$, $\mathrm{t\bar{t}}$+HF
- **Simultaneous** fit of $\mathrm{t\bar{t}H(H \to c\bar{c})}$, $\mathrm{t\bar{t}H(H \to b\bar{b})}$, $\mathrm{t\bar{t}Z(Z \to b\bar{b}/c\bar{c})}$
  - $\mathrm{t\bar{t}}$+cc/bb normalised in dedicated regions
  - $\mathrm{t\bar{t}Z(Z \to c\bar{c})}$ as a standard candle: $\mu = 1.0 \pm 0.8$
- By-product: $\mathrm{t\bar{t}H(H \to b\bar{b})}$ $\mu = 0.91^{+0.26}_{-0.22}$, 4.4σ (4.5 exp.)



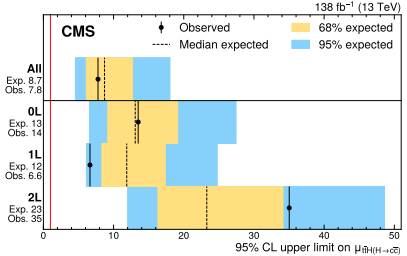
b, c, light jets in the ParticleNet (b vs. light, c vs. light) plane and the 2D tagging categories

$\mathrm{t\bar{t}H(H \to c\bar{c})}$: results

$\mathrm{t\bar{t}H(H \to c\bar{c})}$, 138 fb^{-1}

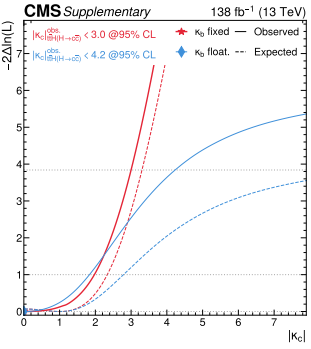
$\mu_{\mathrm{t\bar{t}H(cc)}} < 7.8\,(8.7)$
$\sigma\mathcal{B} < 0.11\,(0.13)$ pb
$|\kappa_\mathrm{c}| < 3.0\,(3.3)$, $\kappa_\mathrm{b} = 1$
$|\kappa_\mathrm{c}| < 4.2$ obs., κ_b floating

- Expected sensitivity comparable to $\mathrm{VH(H \to c\bar{c})}$
- Simultaneous bb/cc fit constrains κ_b and κ_c together



$\mu_{\mathrm{t\bar{t}H(cc)}}$ limits per channel

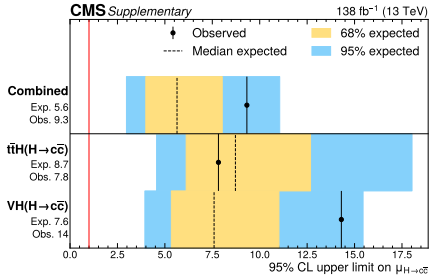
PRL 136 (2026) 011801, CMS-HIG-24-018

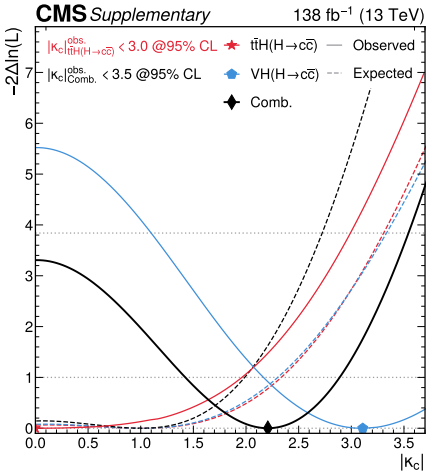


$|\kappa_\mathrm{c}|$ scan, κ_b fixed / floating

Combination of ttH and VH: tightest constraint on κ_c

PRL 136 (2026) 011801, CMS-HIG-24-018





ttH + VH

$\mu_{H\to c\bar{c}} < 9.3$
(5.6 exp.)

$|\kappa_c| < 3.5$
(2.7 exp.)

Expected limit improves by ∼20% over the best single channel

Most stringent direct constraint on κ_c to date; expected reach now below the BSM-motivated $|\kappa_c| \sim 3$

Boosted $\mathrm{H} \to \mathrm{c\bar{c}}$

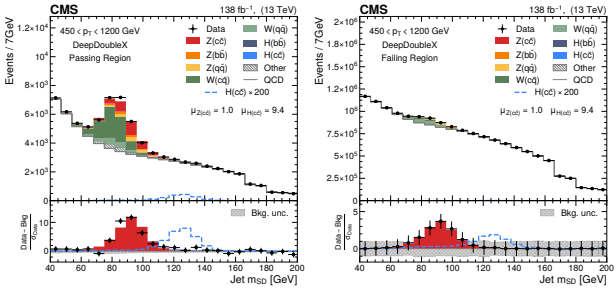
PRL 131 (2023) 041801, CMS-HIG-21-012

- Inclusive H production with $p_{\mathrm{T}} > 450\,\mathrm{GeV}$, dominated by ggH + ISR
- Single large-radius jet, DeepDoubleX $\mathrm{H} \to \mathrm{c\bar{c}}$-tagger
- $\mathrm{Z} \to \mathrm{c\bar{c}}$ observed in the same jets: $\mu_{\mathrm{Z}} = 1.00^{+0.17}_{-0.14}\,(\mathrm{syst}) \pm 0.08\,(\mathrm{theo}) \pm 0.06\,(\mathrm{stat})$

Result, $138\,\mathrm{fb}^{-1}$

$\mu_{\mathrm{H}\to\mathrm{c\bar{c}}} < 47\,(39)$

- Complementary production mode; high-p_{T} region profits most from new mass-decorrelated taggers



Jet mass in the passing region, with fitted $\mathrm{H} \to \mathrm{c\bar{c}}$ and $\mathrm{Z} \to \mathrm{c\bar{c}}$

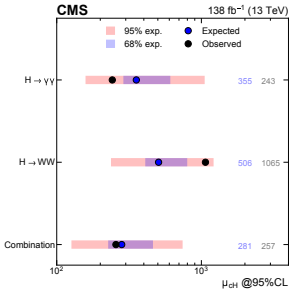
H + charm associated production

- cg→cH: κ_c enters *production*, independent of $H \to c\bar{c}$-decay
- Targets clean decays: **H→ $\gamma\gamma$** and **H→WW→e$\nu\mu\nu$** NEW
- Challenges:
  - small κ_c-dependent rate; H+c also produced via $g \to c\bar{c}$-splitting, independent of y_c
  - soft c-jets; flavour-scheme modelling

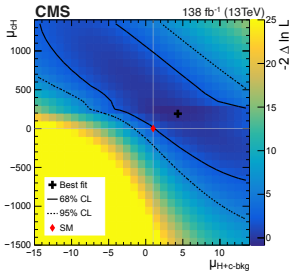
  → T. Bevilacqua's talk, next

95% CL	μ_{cH}	$\lvert\kappa_c\rvert$
H→ $\gamma\gamma$	243 (355)	38.1 (72.5)
H→WW	1065 (506)	211 (95)
Combined	257 (281)	**47 (51)**

JHEP 11 (2025) 060; JHEP 08 (2026) 051



μ_{cH} limits, $\gamma\gamma$ + WW



H→WW: μ_{cH} vs. $\mu_{H+c\ bkg}$

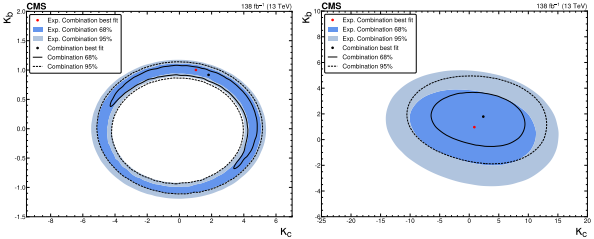
Indirect constraints I: Higgs boson p_{T} spectrum

PLB 792 (2019) 369; JHEP 08 (2023) 040; JHEP 04 (2026) 093

- Light-quark loops and $\mathrm{q\bar{q}} \to \mathrm{H}$ change the **shape** of $p_{\mathrm{T}}^{\mathrm{H}}$; κ_{c}, κ_{b} also change $\mathcal{B}$
- Fits of differential cross sections in $(\kappa_{\mathrm{b}},\, \kappa_{\mathrm{c}})$:

	κ_{c} 95% CL obs. (exp.) $\mathcal{B}(\kappa)$	 $\mathcal{B}$ free
$\gamma\gamma$+4ℓ+bb, 2016	$[-4.9, 4.8]$	$[-33, 38]$
	$[-6.1, 6.0]$	$[-31, 36]$
$\mathrm{H} \to 4\ell$, 138 fb^{-1}	$[-5.3, 5.2]$	$[-20, 23]$
	$[-5.7, 5.7]$	$[-19, 20]$

- **NEW** Run 2 combination of $\gamma\gamma$, ZZ, WW, $\tau\tau$: 2D $(\kappa_{\mathrm{b}},\, \kappa_{\mathrm{c}})$ contours, compatible with SM at 68% CL
- Sensitive to the **sign** of κ_{c}

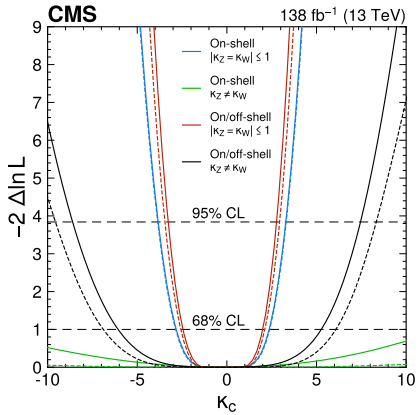


JHEP 04 (2026) 093: $(\kappa_{\mathrm{b}},\, \kappa_{\mathrm{c}})$ from the Run 2 $p_{\mathrm{T}}^{\mathrm{H}}$ combination; left: $\mathcal{B}$ dependent on couplings, right: $\mathcal{B}$ unconstrained

Indirect constraints II: γH and on/off-shell H$\to 4\ell$

- Large light-quark Yukawas raise $q\bar{q}\to$H, γH and change Γ_{H}
- **γH + on-shell H$\to 4\ell$** (PRD 112 (2025) 112001)
  $\kappa_c \in [-3.7, 3.2]$ ($[-3.8, 3.3]$ exp.), other Yukawas floating, $|\kappa_V| \leq 1$
- **NEW Off-shell + on-shell H$\to 4\ell$** (arXiv:2607.23352)
  - Off-shell rate breaks the width degeneracy
    $\Rightarrow$ no assumption on κ_Z, κ_W needed
  - $\kappa_c \in [-8.6, 7.5]$ ($[-9.6, 8.3]$ exp.)
    other light-quark couplings, κ_Z, κ_W profiled
  - With $|\kappa_V| \leq 1$: $\kappa_c \in [-3.3, 2.8]$ ($[-3.5, 2.9]$ exp.)

PRD 112 (2025) 112001; arXiv:2607.23352



κ_c likelihood scans for on-shell and on+off-shell fits, with and without $|\kappa_V| \leq 1$

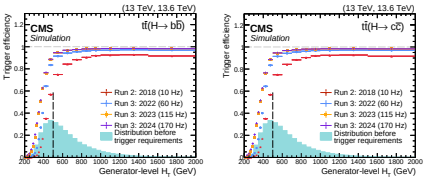
Where we stand: CMS constraints on κ_c

Probe	Reference	κ_c at 95% CL: obs. (exp.)	Main assumptions
$t\bar{t}H + VH(H \to c\bar{c})$	PRL 136 (2026) 011801	$\lvert\kappa_c\rvert < 3.5$ (2.7)	$\kappa_b = 1$, SM-like width
$t\bar{t}H(H \to c\bar{c})$	PRL 136 (2026) 011801	$\lvert\kappa_c\rvert < 3.0$ (3.3)	$\kappa_b = 1$
$VH(H \to c\bar{c})$	PRL 131 (2023) 061801	$1.1 < \lvert\kappa_c\rvert < 5.5$ (3.4)	$\kappa_b = 1$
$\gamma H + H \to 4\ell$	PRD 112 (2025) 112001	$[-3.7, 3.2]$ ($[-3.8, 3.3]$)	$\lvert\kappa_V\rvert \leq 1$
On+off-shell $H \to 4\ell$	arXiv:2607.23352	$[-8.6, 7.5]$ ($[-9.6, 8.3]$)	none on $\kappa_{Z,W}$
p_T^H, $H \to 4\ell$	JHEP 08 (2023) 040	$[-5.3, 5.2]$ ($[-5.7, 5.7]$)	$\mathcal{B}(\kappa_b, \kappa_c)$
cH ($\gamma\gamma$ + WW)	JHEP 08 (2026) 051	$\lvert\kappa_c\rvert < 47$ (51)	
$H \to J/\psi\gamma$	PLB 865 (2025) 139462	$[-166, 208]$	$\kappa_\gamma = 1$

- Direct $H \to c\bar{c}$–searches lead; indirect probes add the **sign** of κ_c and weaker model assumptions
- Several probes now reach $|\kappa_c| \sim 3$, where flavour-compatible BSM scenarios live

Toward Run 3: tools already in place

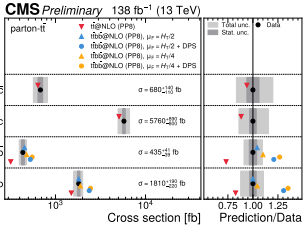
Heavy-flavour triggers



arXiv:2608.18331 (CMS-BTV-25-002): ML b/c tagging at the HLT;

$\mathrm{ttH(H \to c\bar{c})}$ trigger efficiency, Run 2 vs. 2022–24

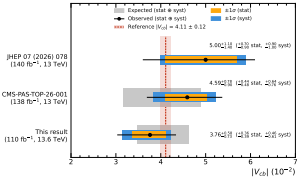
$\mathrm{tt}+\mathrm{c\bar{c}}$ background measured



CMS-PAS-TOP-26-003 **NEW**:

$\sigma_{\mathrm{ttcc}} = 680^{+130}_{-100}\,(\mathrm{syst}) \pm 40\,(\mathrm{stat})\,\mathrm{fb}$

c-tagging in Run 3 data



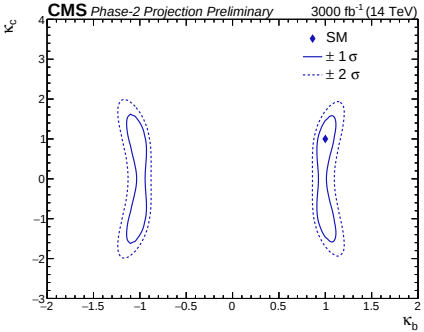
CMS-PAS-TOP-26-002 **NEW**: $|V_{\mathrm{cb}}|$ from W→cb in tt, 110 fb^{-1} at 13.6 TeV, with UParT

- Run 3 at 13.6 TeV: Run 3 dataset already larger than Run 2, new taggers (UParT, GloParT), parking and scouting streams for hadronic final states
- Run 3 analyses are under way in all channels

Prospects at the HL-LHC

- ATLAS + CMS input to the European Strategy update
  (arXiv:2504.00672, CMS-HIG-25-002), $3\,\mathrm{ab}^{-1}$ per experiment:
  - VH($\mathrm{H} \to \mathrm{c\bar{c}}$): $\approx 1.6\sigma$
  - $|\kappa_\mathrm{c}| < 1.5$ at 95% CL, 30% better than previous projections
- History: expected $\mu_\mathrm{H} \to \mathrm{c\bar{c}}$ limit improved from $\sim$37 (2016 data) to 5.6 (Run 2 VH+$\mathrm{t\bar{t}H}$)
  far faster than luminosity alone: taggers, new channels, analysis techniques
- Further gains: Run 3 data, timing and track-trigger upgrades, ML tagging; theory modelling and tagger calibration become limiting

Is $\mathrm{H} \to \mathrm{c\bar{c}}$ evidence reachable at the HL-LHC?
Determining $|\kappa_\mathrm{c}|$ at $\mathcal{O}(1)$ is.



CMS-HIG-21-008 (supplementary): expected $(\kappa_\mathrm{b}, \kappa_\mathrm{c})$ contours at $3000\ \mathrm{fb}^{-1}$, 14 TeV, extrapolated from the Run 2 VH($\mathrm{H} \to \mathrm{c\bar{c}}$) analysis

Summary

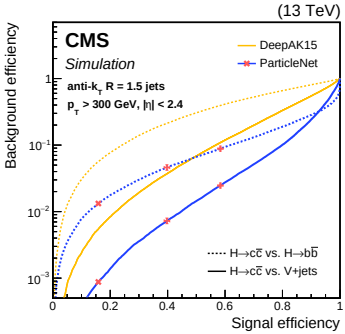
- The charm Yukawa is the most accessible 2nd-generation quark coupling, and CMS probes it in many complementary ways
- Direct searches: $\mathrm{t\bar{t}H}(\mathrm{H} \to \mathrm{c\bar{c}})$ added to VH and boosted $\mathrm{H} \to \mathrm{c\bar{c}}$
  $\Rightarrow |\kappa_\mathrm{c}| < 3.5$ **(2.7 exp.)**, the most stringent constraint to date
- H+c production with $\gamma\gamma$ and WW: $|\kappa_\mathrm{c}| < 47$ (51 exp.)
- Indirect probes (p_T^H, γH, on/off-shell H$\to 4\ell$, H$\to$J/$\psi\gamma$) are sensitive to the sign of κ_c and relax model assumptions
- Sensitivity is now reaching the BSM-motivated region $|\kappa_\mathrm{c}| \sim 3$
- Run 3 tools (HLT tagging, UParT, $\mathrm{t\bar{t}}$+HF measurements) are in place; HL-LHC projects $|\kappa_\mathrm{c}| < 1.5$

Thank you!

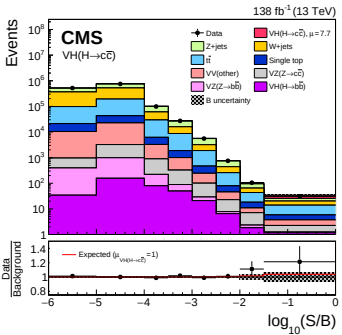
Backup

VH($\mathrm{H} \to \mathrm{c\bar{c}}$): event categories and sensitivity

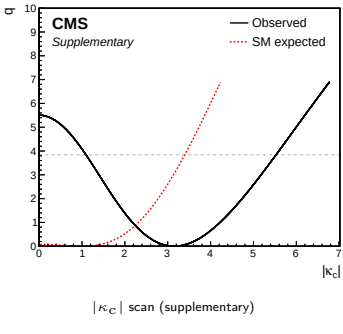
PRL 131 (2023) 061801



ParticleNet merged-jet $\mathrm{H} \to \mathrm{c\bar{c}}$-tagging



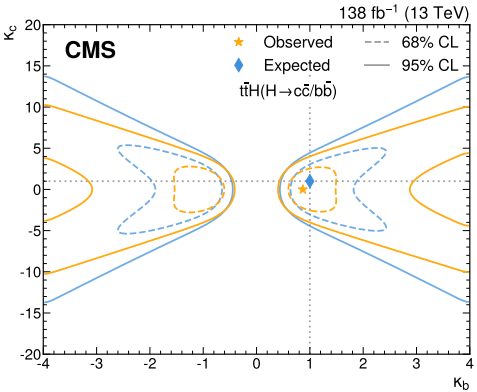
Events vs. $\log_{10}(S/B)$



Best fit $\mu = 7.7^{+3.8}_{-3.5}$; merged $\mu = 8.7^{+4.6}_{-4.0}$, resolved $\mu = -9.5 \pm 9.6$. $\sigma\mathcal{B} < 0.94$ (0.50) pb.

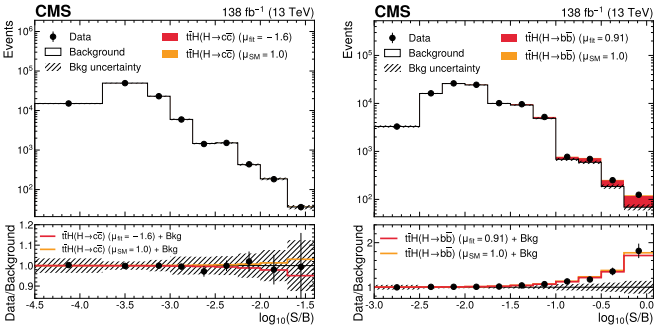
$\mathrm{t\bar{t}H(H \to c\bar{c}/H \to b\bar{b})}$: (κ_b, κ_c) plane

PRL 136 (2026) 011801

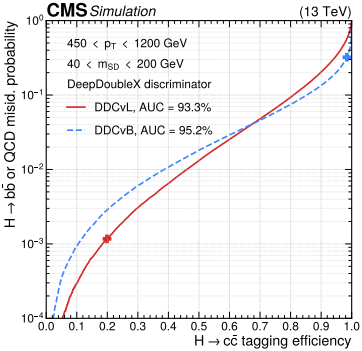


- Simultaneous $\mathrm{t\bar{t}H(H \to b\bar{b})}$ + $\mathrm{t\bar{t}H(H \to c\bar{c})}$ fit constrains both Yukawas
- Signal strengths: $\mathrm{t\bar{t}H}$(bb) $0.91^{+0.26}_{-0.22}$; $\mathrm{t\bar{t}H}$(cc) -1.6 ± 4.5; $\mathrm{t\bar{t}Z}$(bb) $1.47^{+0.45}_{-0.41}$; $\mathrm{t\bar{t}Z}$(cc) $1.02^{+0.79}_{-0.84}$

$\mathrm{t\bar{t}H(H \to c\bar{c})}$ and boosted $\mathrm{H \to c\bar{c}}$: more figures

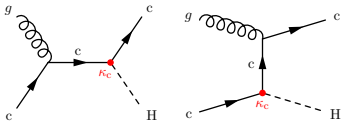


PRL 136 (2026) 011801: event yields vs. $\log_{10}(S/B)$

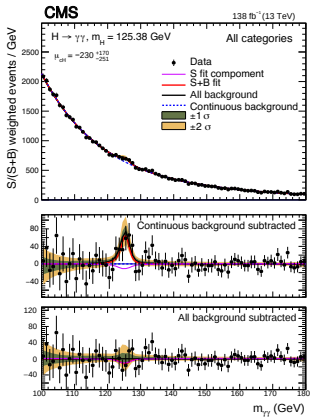


PRL 131 (2023) 041801: DeepDoubleX performance

H+c: H→ $\gamma\gamma$ and H→WW

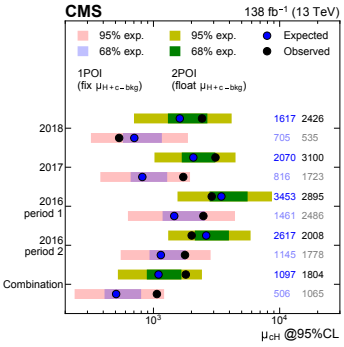


LO diagrams for cH production



cH($\gamma\gamma$): $m_{\gamma\gamma}$, all categories

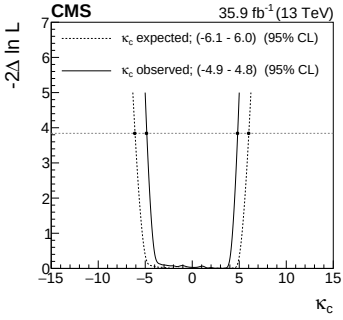
JHEP 11 (2025) 060; JHEP 08 (2026) 051



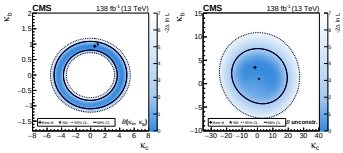
cH(WW): μ_{cH} limits per period, 1 and 2 POI

cH(WW) 2-POI fit (H+c background floating): $\mu_{cH} < 1804$ (1097 exp.).

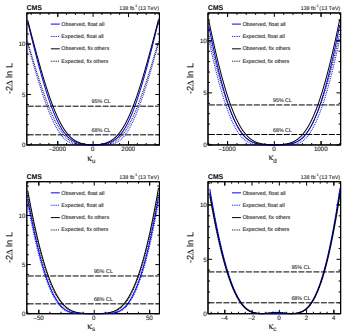
Indirect: $p_{\mathrm{T}}^{\mathrm{H}}$ and γH details



PLB 792 (2019) 369: κ_{c} scan, $\mathcal{B}(\kappa)$



JHEP 08 (2023) 040: H$\rightarrow 4\ell$ (κ_{b}, κ_{c})

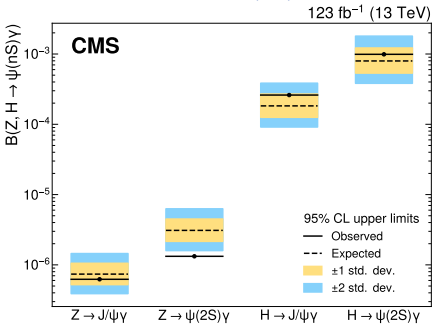


PRD 112 (2025) 112001: light-quark κ scans

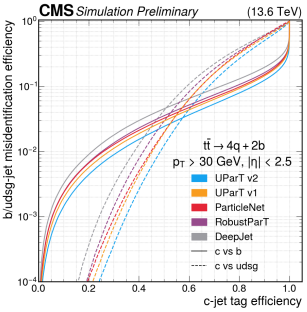
Rare exclusive decays: $H \to J/\psi\gamma$, $\psi(2S)\gamma$

- Direct ($\propto \kappa_c$) and indirect ($H \to \gamma\gamma^* \to \gamma\, J/\psi$) amplitudes interfere $\Rightarrow$ sign sensitivity
- $123\,\mathrm{fb}^{-1}$: $\mathcal{B}(H \to J/\psi\gamma) < 2.6 \times 10^{-4}$, $\mathcal{B}(H \to \psi(2S)\gamma) < 9.9 \times 10^{-4}$
- $\kappa_c/\kappa_\gamma \in (-157, 199)$; with $\kappa_\gamma = 1$: $\kappa_c \in (-166, 208)$

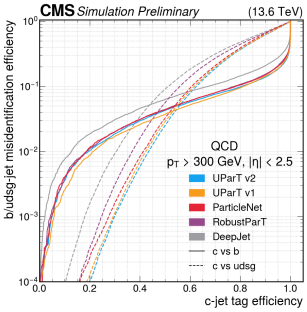
PLB 865 (2025) 139462, CMS-SMP-22-012



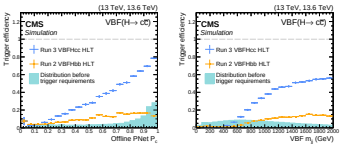
Run 3 jet-flavour tagging performance



CMS-DP-2025-081: c-tagging ROC, $t\bar{t}$



CMS-DP-2025-081: c-tagging ROC, QCD



arXiv:2608.18331: VBF H$\to c\bar{c}$ trigger efficiency

References

- VH($\mathrm{H} \to \mathrm{c\bar{c}}$): CMS, PRL 131 (2023) 061801, arXiv:2205.05550 (HIG-21-008)
- $\mathrm{t\bar{t}H}(\mathrm{H} \to \mathrm{c\bar{c}})$ + VH combination: CMS, PRL 136 (2026) 011801, arXiv:2509.22535 (HIG-24-018)
- Boosted $\mathrm{H} \to \mathrm{c\bar{c}}$: CMS, PRL 131 (2023) 041801, arXiv:2211.14181 (HIG-21-012)
- cH(H$\to \gamma\gamma$): CMS, JHEP 11 (2025) 060, arXiv:2503.08797 (HIG-23-010)
- cH(H$\to$WW) and combination: CMS, JHEP 08 (2026) 051, arXiv:2508.14988 (HIG-24-009)
- $p_{\mathrm{T}}^{\mathrm{H}}$ combination 2016: CMS, PLB 792 (2019) 369, arXiv:1812.06504 (HIG-17-028)
- H$\to 4\ell$ differential: CMS, JHEP 08 (2023) 040, arXiv:2305.07532 (HIG-21-009)
- Differential combination Run 2: CMS, JHEP 04 (2026) 093, arXiv:2504.13081 (HIG-23-013)
- γH + H$\to 4\ell$ light Yukawas: CMS, PRD 112 (2025) 112001, arXiv:2502.05665 (HIG-23-011)
- Off-shell + on-shell H$\to 4\ell$: CMS, arXiv:2607.23352, submitted to PRD (HIG-25-003)
- H$\to$J/$\psi\gamma$: CMS, PLB 865 (2025) 139462, arXiv:2411.15000 (SMP-22-012)
- HLT flavour tagging: CMS, arXiv:2608.18331 (BTV-25-002); tagging: CMS-DP-2025-081
- $\mathrm{t\bar{t}}$+b/c: CMS-PAS-TOP-26-003; $|V_{\mathrm{cb}}|$ at 13.6 TeV: CMS-PAS-TOP-26-002
- HL-LHC: ATLAS and CMS, arXiv:2504.00672 (HIG-25-002)
- BSM: arXiv:2410.05236